AWS CERTIFIED SOLUTIONS ARCHITECT GUIDE

AWS Networking & ELB Deep Dive

Comprehensive Technical Study Guide for VPC, Subnets, Gateways, Route Tables & Load Balancers

CORE TOPICS COVERED

- ✔ Amazon Virtual Private Cloud (VPC) Architecture
- ✔ Public & Private Subnet Design Strategies
- ✔ Internet Gateways (IGW) & Egress-Only IGW
- ✔ NAT Gateways vs NAT Instances
- ✔ Route Tables, Subnet Associations & Routes
- ✔ Elastic Load Balancing (ALB, NLB, GWLB, CLB)

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1. Amazon Virtual Private Cloud (VPC)

An **Amazon Virtual Private Cloud (VPC)** is a logically isolated virtual network dedicated to your AWS account. It closely resembles a traditional network that you'd operate in your own data center, with the benefits of using the scalable infrastructure of AWS.

1.1 Key VPC Concepts & Fundamentals

- **CIDR Block Allocation:** When creating a VPC, you specify an IPv4 Classless Inter-Domain Routing (CIDR) block (e.g., `10.0.0.0/16`). The block size can range from `/16` (65,536 IPs) to `/28` (16 IPs).
- **Region Scope:** A VPC spans all Availability Zones (AZs) within a specific AWS Region. It does not span multiple regions.
- **Default vs Custom VPC:** Every AWS account comes with a Default VPC pre-configured with public subnets in each AZ, an attached Internet Gateway, and public IPv4 addressing enabled. Custom VPCs provide complete control over subnetting, IP assignment, and routing logic.

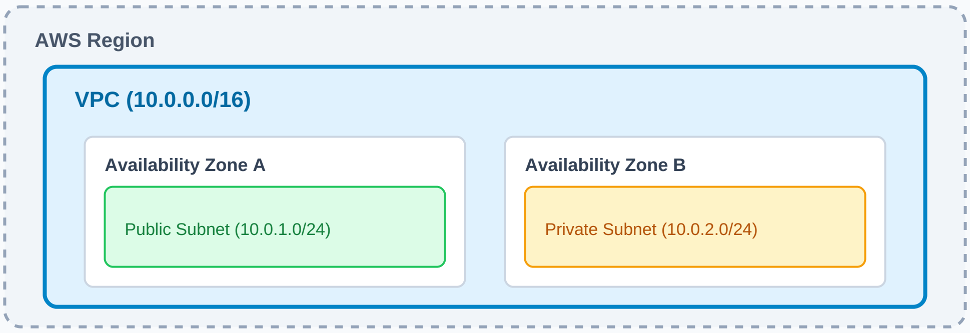

FIGURE 1: HIGH-LEVEL AWS VPC ARCHITECTURE WITH MULTI-AZ SUBNETS

2. Subnets (Public & Private)

A **Subnet** is a range of IP addresses in your VPC. You can launch AWS resources, such as Amazon EC2 instances, into specific subnets. Subnets reside entirely within a single Availability Zone.

2.1 AWS Reserved IP Addresses

In every subnet created within AWS, **5 IP addresses** are reserved by AWS and cannot be assigned to host devices. For example, in a `10.0.0.0/24` subnet:

IP Address	Reserved Purpose / Description
`10.0.0.0`	Network Address (Base network address).
`10.0.0.1`	Reserved by AWS for the VPC Router.
`10.0.0.2`	Reserved by AWS for DNS server mapping.
`10.0.0.3`	Reserved by AWS for future hardware/feature expansions.
`10.0.0.255`	Network Broadcast Address (AWS does not support broadcast in VPCs).

2.2 Public vs Private Subnets

- **Public Subnet:** Contains a route in its route table pointing to an *Internet Gateway (IGW)*. Resources in a public subnet can have public IP addresses and communicate directly with the internet.
- **Private Subnet:** Does not have a direct route to an Internet Gateway. Outbound internet connectivity requires a *NAT Gateway* or *NAT Instance* located in a public subnet.

Author's Exam Tip — Prabu Thiagamani

Always calculate usable IP addresses properly on exams! A `/24` subnet has 256 total IPs, but only **251 usable IPs** due to the 5 AWS reserved addresses.

3. Internet Gateway (IGW) vs NAT Devices

3.1 Internet Gateway (IGW)

An **Internet Gateway** is a horizontally scaled, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet. It imposes no availability risks or bandwidth constraints on your network traffic.

- Performs Network Address Translation (NAT) for instances with public IPv4 addresses.
- Serves as a target in VPC route tables for internet-bound traffic (`0.0.0.0/0`).

3.2 NAT Gateway vs NAT Instance

To enable instances in a private subnet to connect to the internet (e.g., for software updates) while preventing the internet from initiating connections with those instances, you use a Network Address Translation (NAT) device.

Feature / Capability	Managed NAT Gateway	Self-Managed NAT Instance
Managed Service	Fully managed by AWS (High availability built-in)	Self-managed EC2 instance (User manages OS/ patches)
Bandwidth	Scales automatically up to 100 Gbps	Depends on EC2 instance type bandwidth capacity
Maintenance	Zero maintenance required	Requires OS updates, security patching, monitoring
High Availability	Redundant within AZ; deploy across AZs for regional HA	Requires custom failover scripts / auto-scaling groups
Security Groups	Cannot associate Security Groups directly	Can associate Security Groups directly
Port Forwarding	Not supported	Supported via custom iptables rules

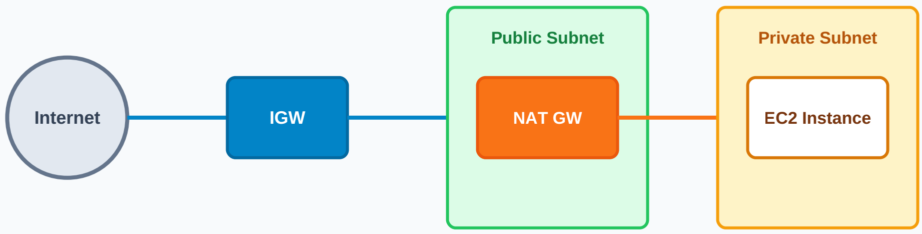

FIGURE 2: OUTBOUND TRAFFIC FLOW FROM PRIVATE SUBNET VIA NAT GATEWAY AND IGW

4. Route Tables

A **Route Table** contains a set of rules (called routes) that determine where network traffic from your subnet or gateway is directed.

- **Main Route Table:** The route table that automatically comes with your VPC. It controls the routing for all subnets that are not explicitly associated with any other route table.
- **Custom Route Table:** A route table created manually for specific subnets to override main route behavior.
- **Subnet Association:** Each subnet must be associated with a route table. If not explicitly associated, it inherits the Main Route Table.

4.1 Sample Route Table Configurations

Public Subnet Route Table Example:

Destination CIDR	Target	Description / Purpose
`10.0.0.0/16`	`local`	Internal VPC communications (Default, non-deletable route).
`0.0.0.0/0`	`igw-0a1b2c3d4e5f`	All internet-bound traffic sent directly to Internet Gateway.

Private Subnet Route Table Example:

Destination CIDR	Target	Description / Purpose
`10.0.0.0/16`	`local`	Internal VPC communications.
`0.0.0.0/0`	`nat-0123456789abcdef`	All outbound internet traffic routed safely through NAT Gateway.

AWS LOAD BALANCING DEEP DIVE *Author: Prabu Thiagamani*

5. Elastic Load Balancing (ELB) Types

Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets, such as EC2 instances, containers, IP addresses, and Lambda functions. AWS provides 4 distinct types of load balancers tailored for specific use cases.

5.1 Comprehensive Comparison of ELB Types

Feature / Dimension	Application Load Balancer (ALB)	Network Load Balancer (NLB)	Gateway Load Balancer (GWLB)	Classic Load Balancer (CLB)
OSI Model Layer	Layer 7 (HTTP / HTTPS / gRPC)	Layer 4 (TCP / UDP / TLS)	Layer 3 (IP Packets)	Layer 4 / Layer 7 (Legacy)
Performance / Latency	High performance (~ms latency)	Ultra-high performance (<1 ms latency)	Ultra-low latency with GENEVE encaps.	Standard performance
Static IP Support	No (Provides DNS name)	Yes (Static IP per AZ + Elastic IP)	Yes (Static IP per AZ)	No (Provides DNS name)
Routing Capabilities	Path, Host, Query String, Header routing	Flow hash based on Layer 4 parameters	Routes traffic to 3rd party virtual appliances	Basic round-robin routing
Target Types	EC2, IP, Lambda, Containers	EC2, IP, ALB	IP, EC2 Instances	EC2 Instances (Classic)
TLS Offloading / Certs	Supported (SNI support)	Supported (Zero CPU overhead on target)	N/A (Pass-through mode)	Supported (Single certificate)

Feature / Dimension	Application Load Balancer (ALB)	Network Load Balancer (NLB)	Gateway Load Balancer (GWLB)	Classic Load Balancer (CLB)
Primary Use Case	Modern Microservices & Web Apps	High-throughput, extreme scale & gaming	Inline Security & Firewall Appliances	Legacy workloads (Deprecation recommended)

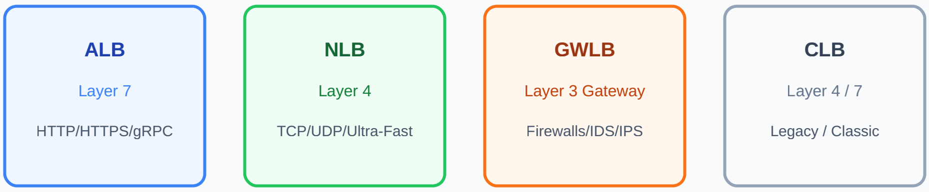

FIGURE 3: OVERVIEW OF AWS ELASTIC LOAD BALANCER (ELB) SUITE

5.2 Key Features & Advanced Routing Rules

- **Sticky Sessions (Session Affinity):** Ensures requests from the same client are directed to the same target using HTTP cookies (Available on ALB & CLB).
- **Cross-Zone Load Balancing:** Distributes traffic evenly across all registered targets in all enabled Availability Zones rather than just within the local AZ. Default ON for ALB, optional/disabled by default for NLB.
- **Health Checks:** Periodically sends requests to registered targets to verify health. Unhealthy targets are automatically removed from routing until restored.

Summary Checklist for Network Design

- Place web servers/public load balancers in **Public Subnets**.
- Place application servers, databases, and internal backends in **Private Subnets**.
- Ensure high availability by deploying resources across at least **2 Availability Zones** with corresponding NAT Gateways.